

CSD2161 Network

Assignment 5

Group: A20

Members:

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Design Introduction

Overview

"Tank Wars" is a top down 2D 4 player arena shooter with LAN multiplayer. Players control an immersively controlled tank in a free for all deathmatch style match with the first player reaching 5 kills winning the match.

With the persistent server side EXP system, players upon getting 10kills can upgrade their tank gun.

A global top 5 kills leaderboard in the waiting room breeds competition in the player to get better

A chat function in the match to add to the multiplayer experience

Architecture

The game uses a strict Server Authoritative Model that runs a 60 tick server game loop that handles collision, input processing, projectile TTL and data persistency

Total kills per player are stored in a plain text file "player_database.txt" which players can spend 10kills to upgrade their tank gun.

TCP is used for client connecting to the server, toggling of ready in waiting room, in game chat function, buying of upgrades and the cheat win key for testing.

UDP is used for input and all game packets

The client does not simulate any physics locally, the server handles all physics, movement, shooting and kill tracking. Thus all the client does is render, send inputs by the player and display what the server says

Disruption handling & Robustness

If a client TCP socket closes or disconnects unexpectedly, the server will flag the player as disconnected, update their lifetime kills to the database and resets the player's coordinates off screen so the remaining active players can continue the match uninterrupted

The client side has a DisconnectFromServer function that acts as the cleanup functions that safely joins background network threads, wipes lobby memory and returns the application to the main menu to prevent crashes or memory leaks

If client joins back from a unexpected disconnection the player will resume from main menu and can continue to play the game as expected

Sequence

Channel	Direction	Packet	Purpose
UDP	Client to Server	InputPacket	Key presses and aiming angle
UDP	Server to Client (broadcast)	GameStateHeader + PlayerState[] + ProjectileState[]	Authoritative world state
TCP	Client to Server	REQ_JOIN, REQ_TOGGLE_READY, REQ_CHAT, REQ_BUY_UPGRADE, REQ_CHEAT_WIN	Reliable control events
TCP	Server to Client (broadcast)	REQ_CHAT + length + message	In game chat

Feature	Initiator	Channel	Content
Player Movement	Client	UDP	InputPacket every frame
Game State broadcast	Server	UDP	GameStateHeader + players + projectiles
Ready Toggle	Client	TCP	REQ_TOGGLE_READY (1byte)
Chat	Client	TCP	REQ_CHAT + length + message
Leaderboard	Client request, Server responds	TCP	REQ_LEADERBOARD + LeaderboardEntry[]

Individual Contribution

Lai Jun Jie Clement, 2403344, junjieclement.lai	35%
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Lee Hwee Min, 2403441, l.hweemin	30%

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